

Disciplina: **ENE0025 – Protocolos de Transporte e Roteamento**

Curso: **Engenharia de Redes de Comunicação**

Instituição: **Universidade de Brasília (UnB)**

Departamento: **Engenharia Elétrica**

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Relatório do Laboratório 7

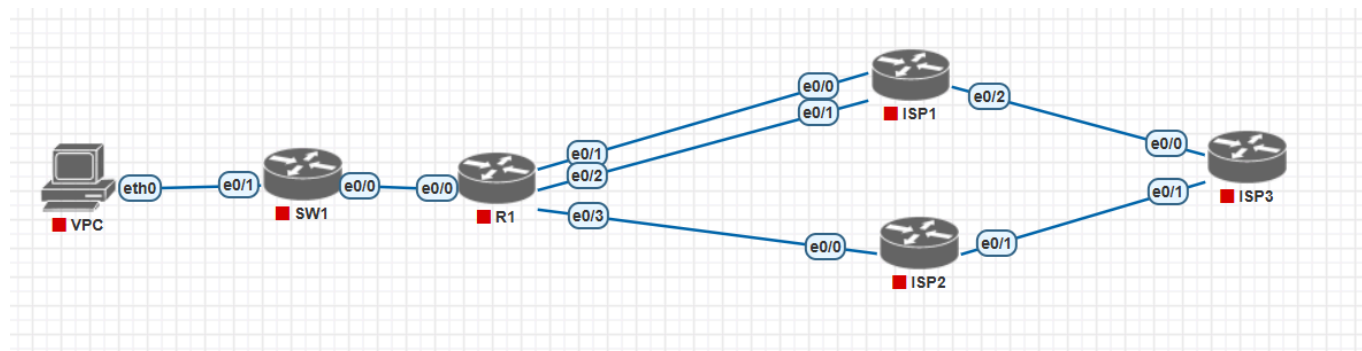
Identificação

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Objetivo

Como este laboratório é uma continuação do laboratório 6, o objetivo agora é configurar os roteadores ISP1, ISP2 e ISP3 para permitir o funcionamento completo do cenário de BGP externo.

Topologia do Laboratório



Considerando a topologia do laboratório 06:

- **R1** pertence ao **AS 1000**
- **ISP1** pertence ao **AS 100**
- **ISP2** pertence ao **AS 200**
- **ISP3** pertence ao **AS 300**

Os enlaces considerados são:

- **R1 ↔ ISP1**
 - rede **10.1.0.0/30**
 - rede **10.1.0.4/30**
- **R1 ↔ ISP2**

- rede **10.2.0.0/30**
- **ISP1 ↔ ISP3**
 - rede **191.1.0.0/30**
- **ISP2 ↔ ISP3**
 - rede **191.2.0.0/30**

Também será utilizada a loopback:

- **ISP1: 10.10.10.10/32**

E os prefixos externos anunciados por **ISP3**:

- 181.0.0.0/8
- 182.0.0.0/8
- 183.0.0.0/8
- 184.0.0.0/8
- 185.0.0.0/8

Procedimentos

Configuração do ISP3

O **ISP3** representa a rede externa com os prefixos de teste.

Interfaces do ISP3

```
ISP3> enable
ISP3# configure terminal
ISP3(config)# no ip domain lookup
ISP3(config)# interface e0/0
ISP3(config-if)# ip address 191.1.0.2 255.255.255.252
ISP3(config-if)# no shutdown
ISP3(config-if)# interface e0/1
ISP3(config-if)# ip address 191.2.0.2 255.255.255.252
ISP3(config-if)# no shutdown
ISP3(config-if)# interface loopback 1
ISP3(config-if)# ip address 181.0.0.1 255.0.0.0
ISP3(config-if)# interface loopback 2
ISP3(config-if)# ip address 182.0.0.1 255.0.0.0
ISP3(config-if)# interface loopback 3
ISP3(config-if)# ip address 183.0.0.1 255.0.0.0
ISP3(config-if)# interface loopback 4
ISP3(config-if)# ip address 184.0.0.1 255.0.0.0
ISP3(config-if)# interface loopback 5
```

```
ISP3(config-if)# ip address 185.0.0.1 255.0.0.0
ISP3(config-if)# end
```

BGP no ISP3

```
ISP3> enable

ISP3# configure terminal

ISP3(config)# router bgp 300

ISP3(config-router)# neighbor 191.1.0.1 remote-as 100

ISP3(config-router)# neighbor 191.1.0.1 password SENHA

ISP3(config-router)# neighbor 191.2.0.1 remote-as 200

ISP3(config-router)# neighbor 191.2.0.1 password SENHA

ISP3(config-router)# network 181.0.0.0 mask 255.0.0.0

ISP3(config-router)# network 182.0.0.0 mask 255.0.0.0

ISP3(config-router)# network 183.0.0.0 mask 255.0.0.0

ISP3(config-router)# network 184.0.0.0 mask 255.0.0.0

ISP3(config-router)# network 185.0.0.0 mask 255.0.0.0

ISP3(config-router)# end
```

Configuração do ISP1

O **ISP1** possui dois enlaces físicos com o **R1**, além de uma loopback usada como vizinho BGP.

Interfaces do ISP1

```
ISP1> enable

ISP1# configure terminal

ISP1(config)# no ip domain lookup

ISP1(config)# interface loopback 0

ISP1(config-if)# ip address 10.10.10.10 255.255.255.255

ISP1(config-if)# no shutdown

ISP1(config-if)# interface e0/0

ISP1(config-if)# ip address 10.1.0.2 255.255.255.252

ISP1(config-if)# no shutdown
```

```
ISP1(config-if)# interface e0/1
ISP1(config-if)# ip address 10.1.0.6 255.255.255.252
ISP1(config-if)# no shutdown
ISP1(config-if)# interface e0/2
ISP1(config-if)# ip address 191.1.0.1 255.255.255.252
ISP1(config-if)# no shutdown
ISP1(config-if)# end
```

BGP no ISP1

```
ISP1> enable
ISP1# configure terminal
ISP1(config)# router bgp 100
ISP1(config-router)# neighbor 11.11.11.11 remote-as 1000
ISP1(config-router)# neighbor 11.11.11.11 password SENHA
ISP1(config-router)# neighbor 11.11.11.11 ebgp-multihop 2
ISP1(config-router)# neighbor 11.11.11.11 update-source Loopback0
ISP1(config-router)# neighbor 191.1.0.2 remote-as 300
ISP1(config-router)# neighbor 191.1.0.2 password SENHA
ISP1(config-router)# network 10.10.10.10 mask 255.255.255.255
ISP1(config-router)# exit
ISP1(config)# ip route 11.11.11.11 255.255.255.255 10.1.0.1
ISP1(config)# ip route 11.11.11.11 255.255.255.255 10.1.0.5
ISP1(config)# end
```

Configuração do ISP2

O **ISP2** possui um enlace direto com o **R1** e um enlace com o **ISP3**.

Interfaces do ISP2

```
ISP2> enable
ISP2# configure terminal
ISP2(config)# no ip domain lookup
ISP2(config)# interface e0/0
ISP2(config-if)# ip address 10.2.0.2 255.255.255.252
ISP2(config-if)# no shutdown
ISP2(config-if)# interface e0/1
ISP2(config-if)# ip address 191.2.0.1 255.255.255.252
ISP2(config-if)# no shutdown
ISP2(config-if)# end
```

BGP no ISP2

```
ISP2> enable

ISP2# configure terminal

ISP2(config)# router bgp 200

ISP2(config-router)# neighbor 10.2.0.1 remote-as 1000

ISP2(config-router)# neighbor 10.2.0.1 password SENHA

ISP2(config-router)# neighbor 191.2.0.2 remote-as 300

ISP2(config-router)# neighbor 191.2.0.2 password SENHA

ISP2(config-router)# end
```

Verificação

R1

- Router# show ip bgp summary

```
R1#show ip bgp summary
BGP router identifier 11.11.11.11, local AS number 1000
BGP table version is 8, main routing table version 8
7 network entries using 1008 bytes of memory
12 path entries using 1008 bytes of memory
4/3 BGP path/bestpath attribute entries using 640 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2728 total bytes of memory
BGP activity 7/0 prefixes, 12/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
10.2.0.2      4    200     45     46      8    0    0 00:36:54      5
10.10.10.10   4    100     44     45      8    0    0 00:36:57      6
```

- Router# show ip bgp

```

R1#show ip bgp
BGP table version is 8, local router ID is 11.11.11.11
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
r>  10.10.10.10/32   10.10.10.10             0           0 100 i
*   181.0.0.0/8      10.10.10.10             0           0 100 300 i
*>  10.2.0.2         10.2.0.2                0           0 200 300 i
*   182.0.0.0/8      10.10.10.10             0           0 100 300 i
*>  10.2.0.2         10.2.0.2                0           0 200 300 i
*   183.0.0.0/8      10.10.10.10             0           0 100 300 i
*>  10.2.0.2         10.2.0.2                0           0 200 300 i
*   184.0.0.0/8      10.10.10.10             0           0 100 300 i
*>  10.2.0.2         10.2.0.2                0           0 200 300 i
*   185.0.0.0/8      10.10.10.10             0           0 100 300 i
*>  10.2.0.2         10.2.0.2                0           0 200 300 i
*>  200.18.245.64/27 0.0.0.0                 0          32768 i

```

- Router# show ip route

```

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C    10.1.0.0/30 is directly connected, Ethernet0/1
L    10.1.0.1/32 is directly connected, Ethernet0/1
C    10.1.0.4/30 is directly connected, Ethernet0/2
L    10.1.0.5/32 is directly connected, Ethernet0/2
C    10.2.0.0/30 is directly connected, Ethernet0/3
L    10.2.0.1/32 is directly connected, Ethernet0/3
S    10.10.10.10/32 [1/0] via 10.1.0.6
                      [1/0] via 10.1.0.2
  11.0.0.0/32 is subnetted, 1 subnets
C    11.11.11.11 is directly connected, Loopback1
B    181.0.0.0/8 [20/0] via 10.2.0.2, 00:20:45
B    182.0.0.0/8 [20/0] via 10.2.0.2, 00:20:45
B    183.0.0.0/8 [20/0] via 10.2.0.2, 00:20:45
B    184.0.0.0/8 [20/0] via 10.2.0.2, 00:20:45
B    185.0.0.0/8 [20/0] via 10.2.0.2, 00:20:45
  192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.0.0/24 is directly connected, Ethernet0/0
L    192.168.0.1/32 is directly connected, Ethernet0/0
  200.18.245.0/27 is subnetted, 1 subnets
S    200.18.245.64 is directly connected, Null0

```

- Router# show ip bgp summary

```
Router#show ip bgp summary
BGP router identifier 10.10.10.10, local AS number 100
BGP table version is 13, main routing table version 13
7 network entries using 1008 bytes of memory
13 path entries using 1092 bytes of memory
5/3 BGP path/bestpath attribute entries using 800 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2996 total bytes of memory
BGP activity 7/0 prefixes, 13/0 paths, scan interval 60 secs

Neighbor      V        AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
11.11.11.11    4         1000     42     41      13    0    0 00:34:15        6
191.1.0.2      4         300      11     11      13    0    0 00:01:05        6
```

- Router# show ip bgp

```
Router#show ip bgp
BGP table version is 13, local router ID is 10.10.10.10
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
*>  10.10.10.10/32    0.0.0.0          0         32768  i
*>  181.0.0.0/8       191.1.0.2         0           0 300  i
*                   11.11.11.11         0       1000 200 300  i
*>  182.0.0.0/8       191.1.0.2         0           0 300  i
*                   11.11.11.11         0       1000 200 300  i
*>  183.0.0.0/8       191.1.0.2         0           0 300  i
*                   11.11.11.11         0       1000 200 300  i
*>  184.0.0.0/8       191.1.0.2         0           0 300  i
*                   11.11.11.11         0       1000 200 300  i
*>  185.0.0.0/8       191.1.0.2         0           0 300  i
*                   11.11.11.11         0       1000 200 300  i
*   200.18.245.64/27  191.1.0.2         0           0 300 200 1000  i
*>                   11.11.11.11         0       1000  i
```

- Router# show ip route

```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

```

Gateway of last resort is not set

```

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.1.0.0/30 is directly connected, Ethernet0/0
L    10.1.0.2/32 is directly connected, Ethernet0/0
C    10.1.0.4/30 is directly connected, Ethernet0/1
L    10.1.0.6/32 is directly connected, Ethernet0/1
C    10.10.10.10/32 is directly connected, Loopback0
11.0.0.0/32 is subnetted, 1 subnets
S    11.11.11.11 [1/0] via 10.1.0.5
      [1/0] via 10.1.0.1
B    181.0.0.0/8 [20/0] via 191.1.0.2, 00:10:09
B    182.0.0.0/8 [20/0] via 191.1.0.2, 00:10:09
B    183.0.0.0/8 [20/0] via 191.1.0.2, 00:10:09
B    184.0.0.0/8 [20/0] via 191.1.0.2, 00:10:09
B    185.0.0.0/8 [20/0] via 191.1.0.2, 00:10:09
191.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    191.1.0.0/30 is directly connected, Ethernet0/2
L    191.1.0.1/32 is directly connected, Ethernet0/2
200.18.245.0/27 is subnetted, 1 subnets
B    200.18.245.64 [20/0] via 11.11.11.11, 00:42:08

```

ISP2

- Router# show ip bgp summary

```

Router#show ip bgp summary
BGP router identifier 191.2.0.1, local AS number 200
BGP table version is 8, main routing table version 8
7 network entries using 1008 bytes of memory
8 path entries using 672 bytes of memory
4/3 BGP path/bestpath attribute entries using 640 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2416 total bytes of memory
BGP activity 7/0 prefixes, 8/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
10.2.0.1       4     1000     55     55       8    0    0 00:45:33      2
191.2.0.2       4      300     54     55       8    0    0 00:45:30      6

```

- Router# show ip bgp


```
Router#show ip bgp
BGP table version is 8, local router ID is 191.2.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop          Metric LocPrf Weight Path
*    10.10.10.10/32    191.2.0.2              0 300 100 i
*>         10.2.0.1              0 1000 100 i
*>    181.0.0.0/8      191.2.0.2              0 300 i
*>    182.0.0.0/8      191.2.0.2              0 300 i
*>    183.0.0.0/8      191.2.0.2              0 300 i
*>    184.0.0.0/8      191.2.0.2              0 300 i
*>    185.0.0.0/8      191.2.0.2              0 300 i
*>   200.18.245.64/27 10.2.0.1              0 1000 i
```

- Router# show ip route

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    10.2.0.0/30 is directly connected, Ethernet0/0
L    10.2.0.2/32 is directly connected, Ethernet0/0
B    10.10.10.10/32 [20/0] via 10.2.0.1, 00:47:46
B    181.0.0.0/8 [20/0] via 191.2.0.2, 00:48:01
B    182.0.0.0/8 [20/0] via 191.2.0.2, 00:48:01
B    183.0.0.0/8 [20/0] via 191.2.0.2, 00:48:01
B    184.0.0.0/8 [20/0] via 191.2.0.2, 00:48:01
B    185.0.0.0/8 [20/0] via 191.2.0.2, 00:48:01
  191.2.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    191.2.0.0/30 is directly connected, Ethernet0/1
L    191.2.0.1/32 is directly connected, Ethernet0/1
  200.18.245.0/27 is subnetted, 1 subnets
B    200.18.245.64 [20/0] via 10.2.0.1, 00:48:17
```

ISP3

- Router# show ip bgp summary

```
Router#show ip bgp summary
BGP router identifier 185.0.0.1, local AS number 300
BGP table version is 9, main routing table version 9
7 network entries using 1008 bytes of memory
9 path entries using 756 bytes of memory
5/3 BGP path/bestpath attribute entries using 800 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2660 total bytes of memory
BGP activity 7/0 prefixes, 9/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ  OutQ Up/Down  State/PfxRcd
191.1.0.1      4      100     30     30       9    0    0 00:18:24        2
191.2.0.1      4      200     61     61       9    0    0 00:51:28        2
```

- Router# show ip bgp

```
Router#show ip bgp
BGP table version is 9, local router ID is 185.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
*>  10.10.10.10/32    191.1.0.1             0              0 100 i
*      191.2.0.1             0              0 200 1000 100 i
*>  181.0.0.0/8       0.0.0.0             0             32768 i
*>  182.0.0.0/8       0.0.0.0             0             32768 i
*>  183.0.0.0/8       0.0.0.0             0             32768 i
*>  184.0.0.0/8       0.0.0.0             0             32768 i
*>  185.0.0.0/8       0.0.0.0             0             32768 i
*    200.18.245.64/27 191.1.0.1             0 100 1000 i
*>      191.2.0.1             0 200 1000 i
```

- Router# show ip route

```
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
10.0.0.0/32 is subnetted, 1 subnets
B    10.10.10.10 [20/0] via 191.1.0.1, 00:20:52
C    181.0.0.0/8 is directly connected, Loopback1
    181.0.0.0/32 is subnetted, 1 subnets
L    181.0.0.1 is directly connected, Loopback1
C    182.0.0.0/8 is directly connected, Loopback2
    182.0.0.0/32 is subnetted, 1 subnets
L    182.0.0.1 is directly connected, Loopback2
C    183.0.0.0/8 is directly connected, Loopback3
    183.0.0.0/32 is subnetted, 1 subnets
L    183.0.0.1 is directly connected, Loopback3
C    184.0.0.0/8 is directly connected, Loopback4
    184.0.0.0/32 is subnetted, 1 subnets
L    184.0.0.1 is directly connected, Loopback4
C    185.0.0.0/8 is directly connected, Loopback5
    185.0.0.0/32 is subnetted, 1 subnets
L    185.0.0.1 is directly connected, Loopback5
    191.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    191.1.0.0/30 is directly connected, Ethernet0/0
L    191.1.0.2/32 is directly connected, Ethernet0/0
    191.2.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    191.2.0.0/30 is directly connected, Ethernet0/1
L    191.2.0.2/32 is directly connected, Ethernet0/1
    200.18.245.0/27 is subnetted, 1 subnets
B    200.18.245.64 [20/0] via 191.2.0.1, 00:52:45
```